

Conditioning Regimens are Associated with Distinct Patterns of Microbiota Injury in Allogeneic Hematopoietic Cell Transplantation



Roni Shouval^{1,2}, Nicholas R. Waters³, Antonio L. C. Gomes³, Corrado Zuanelli Brambilla^{1,4,5}, Teng Fei⁶, Sean M. Devlin⁶, Chi L. Nguyen³, Kate A. Markey^{1,7}, Anqi Dai³, John B. Slingerland³, Annelie G. Clurman³, Emily Fontana⁸, Luigi A. Amoretti⁸, Roberta J. Wright⁸, Tobias M. Hohl^{2,8}, Ying Taur^{2,8}, Anthony D. Sung⁹, Daniela Weber¹⁰, Daigo Hashimoto¹¹, Takanori Teshima¹¹, Nelson J. Chao⁹, Ernst Holler¹⁰, Michael Scordo^{1,2}, Sergio A. Giralt^{1,2}, Miguel-Angel Perales^{1,2}, Jonathan U. Peled^{1,2}, and Marcel R.M. van den Brink^{1,2,3}

ABSTRACT

Purpose: The gut microbiota is subject to multiple insults in allogeneic hematopoietic cell transplantation (allo-HCT) recipients. We hypothesized that preparative conditioning regimens contribute to microbiota perturbation in allo-HCT.

Experimental Design: This was a retrospective study that evaluated the relationship between conditioning regimens exposure in 1,188 allo-HCT recipients and the gut microbiome. Stool samples collected from 20 days before transplantation up to 30 days after were profiled using 16S rRNA sequencing. Microbiota injury was quantified by changes in α -diversity.

Results: We identified distinct patterns of microbiota injury that varied by conditioning regimen. Diversity loss was graded into three

levels of conditioning-associated microbiota injury (CMBI) in a multivariable model that included antibiotic exposures. High-intensity regimens, such as total body irradiation (TBI)-thiotepa-cyclophosphamide, were associated with the greatest injury (CMBI III). In contrast, the nonmyeloablative regimen fludarabine-cyclophosphamide with low-dose TBI (Flu/Cy/TBI200) had a low-grade injury (CMBI I). The risk of acute GVHD correlated with CMBI degree. Pretransplant microbial compositions were best preserved with Flu/Cy/TBI200, whereas other regimens were associated with loss of commensal bacteria and expansion of *Enterococcus*.

Conclusions: Our findings support an interaction between conditioning at the regimen level and the extent of microbiota injury.

Introduction

In allogeneic hematopoietic cell transplantation (allo-HCT), conditioning regimens facilitate engraftment and eradicate residual tumor cells. Conditioning regimens are classified into three inten-

sity groups according to their ability to irreversibly ablate hematopoiesis: myeloablative (MAC), reduced intensity (RIC), and nonmyeloablative (NMA; refs. 1, 2). Higher-intensity regimens are considered more toxic to the gastrointestinal tract and are associated with higher risk of GVHD (3). While aggregation of conditioning regimens by intensity groups has proved clinically helpful (1, 2), conditioning drugs act through unique mechanisms and have distinct toxicity profiles. Furthermore, there is considerable intraindividual variability in toxicity patterns and clinical outcomes are incompletely accounted for by clinical features (4).

Intestinal homeostasis is influenced by the intestinal microbiome and the neighboring mucosal tissue, as well as the interaction of microbe-derived factors with host cell populations (5–8). In allo-HCT, the gut mucosa is inflamed and the microbiome markedly altered (9–11); antibiotic exposures partially account for these microbial shifts (12–16). Chemotherapy and radiation may also affect the microbiome (17, 18), but have not been extensively studied in allo-HCT. Furthermore, the microbiome may modulate the sensitivity and metabolism of these interventions (19, 20). In this observational study, we investigated the hypothesis that microbiota injury in allo-HCT is also dependent on conditioning regimens.

Materials and Methods

Patients

Patients that received allo-HCT at Memorial Sloan Kettering Cancer Center (New York, NY) between September 4, 2009 and December 26, 2019 were screened for inclusion. Patients were eligible if they had at least one stool sample collected between days –20 and +30. We excluded patients treated with uncommon

¹Adult Bone Marrow Transplantation Service, Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, New York. ²Department of Medicine, Weill Cornell Medical College, New York, New York. ³Department of Immunology, Sloan Kettering Institute, Memorial Sloan Kettering Cancer Center, New York, New York. ⁴Department of Medical Biotechnologies, University of Siena, Siena, Italy. ⁵Hematology Unit, Department of Oncology, Azienda Ospedaliera Universitaria Senese, Siena, Italy. ⁶Department of Epidemiology and Biostatistics, Memorial Sloan Kettering Cancer Center, New York, New York. ⁷Fred Hutchinson Cancer Research Center, Seattle, Washington. ⁸Infectious Disease Service, Department of Medicine, Memorial Sloan Kettering Cancer Center, New York, New York. ⁹Division of Hematologic Malignancies and Cellular Therapy, Department of Medicine, Duke University Medical Center, Durham, North Carolina. ¹⁰Department of Internal Medicine, University Medical Center, University of Regensburg, Regensburg, Germany. ¹¹Department of Hematology, Hokkaido University Faculty of Medicine, Sapporo, Japan.

J.U. Peled and M.R.M. van den Brink contributed as co-senior authors to this article.

Corresponding Author: Jonathan U. Peled, Memorial Sloan Kettering Cancer Center, 1275 York Ave, New York, NY 10065. Phone: 646-888-2105; E-mail: peledj@mskcc.org

Clin Cancer Res 2023;29:165–73

doi: 10.1158/1078-0432.CCR-22-1254

©2022 American Association for Cancer Research

Translational Relevance

Conditioning regimens have unique patterns of microbiota injury that correlate with GVHD. We propose a novel scale for conditioning-associated microbiota injury. It offers a new axis for assessing conditioning regimens distinct from typical conditioning intensity classification. Our work identifies conditioning strategies that could potentially benefit from microbiota-directed interventions.

conditioning regimens ($n < 40$), those whose samples were collected at the time of a second or third allo-HCT for graft failure or a third allo-HCT, and those who received uncommon GVHD prophylaxis regimens (administered to <10 patients; **Fig. 1A**). Conditioning regimens included were: intravenous busulfan (pharmacokinetics directed) 9.6 to 12.8 mg/kg, melphalan 140 mg/m², and fludarabine 125 mg/m² (**Bu4/Mel/Flu**); fludarabine 160 mg/m² and intravenous busulfan (pharmacokinetics directed) 9.6 to 12.8 mg/kg (**Flu/Bu4**); total body irradiation (TBI) 1,375 cGy, thiopeta 10 mg/kg, and cyclophosphamide 120 mg/kg (**TBI1375/Tt/Cy**); clofarabine 100 mg/m², melphalan 140 mg/m², and thiopeta 10 mg/kg (**Clo/Tt/Mel**); fludarabine 150 mg/m², cyclophosphamide 50 mg/kg, thiopeta 10 mg/kg, and TBI 400 cGy (**Flu/Cy/Tt/TBI400**); melphalan 140 mg/m², thiopeta 5 mg/kg, and fludarabine 160 mg/m² (**Mel/Tt/Flu**); fludarabine 120 mg/m² and melphalan 140 mg/m² (**Flu/Mel**); and fludarabine 125 mg/m², cyclophosphamide 50 mg/kg, and TBI 200 cGy (**Flu/Cy/TBI200**). Conditioning intensity was classified according to standard criteria (**Fig. 1B**; refs. 1, 2).

To validate the association between conditioning intensity and diversity loss, we included an additional cohort of allo-HCT recipients from Duke University Medical Center in Durham, NC; the University Medical Center, University Hospital Regensburg, in Regensburg, Germany; and Hokkaido University Hospital in Sapporo, Japan. The cohort features were previously reported by Peled and colleagues (11).

Patients provided written consent to an Institutional Review Board (IRB)-approved biospecimen-collection protocol.

Stool samples and sequencing

Stool samples were processed by disruption of bacterial cell walls with silica bead-beating, isolation of nucleic acids, and amplifying and sequencing the genomic 16S ribosomal-RNA gene V4-V5 variable region on the Illumina MiSeq platform, as described previously (11). Amplicon sequence variants (ASV) were called using the DADA2 pipeline and mapped to the NCBI 16S rRNA sequence database using BLAST (21). α - and β -diversity were calculated with the Simpson reciprocal index and Bray-Curtis distances, respectively. Laboratory technicians ascertained stool consistency at the time of aliquoting.

The distances in **Fig. 2D** were determined by applying multidimensional scaling (MDS) to the Bray-Curtis distance matrix. For each patient that had samples collected before conditioning, the earliest sample was selected for calculating per-regimen baseline centroids (based on the first two MDS components).

Statistical analysis

Descriptive statistics, including median and interquartile range (IQR) for continuous variables, and percentages for categorical variables, are provided. Acute GVHD was estimated using the cumu-

lative incidence function, with death and relapse considered as competing events. A generalized estimating equation was constructed to assess correlations between clinical covariates, including the timing of sample collection and α -diversity.

Modeling taxonomic abundance with multivariate models

To separate the association of the microbial community composition with conditioning regimens from the influence of the various patient variables, the data were modeled using Microbiome Multivariable Association with Linear Models (MaAsLin2; ref. 22). In short, the dependency structure of the taxonomic composition within each patient was modeled by a patient-specific random effect. Covariates with fixed effects included conditioning regimens and time from HCT to sample collection. An alternative version of the model included additional covariates with fixed effects—antibiotic exposures, diagnoses, GVHD-prophylaxis, previous allo-HCT, age, and graft type. Antibiotic exposure was encoded such that any sample collected the calendar day after the first dose of that antibiotic was considered exposed.

MaAsLin2 was run with a minimum relative abundance threshold 2×10^{-3} (corresponding to the limit of detection given the inclusion criteria of a minimum 1,000 reads per sample), and a prevalence abundance of features being present in greater than or equal to 5% of the samples. Taxa indicated by a “f_” prefix indicate that no annotation was available at the genus level. The base linear model was used on the log-scaled abundances; continuous metadata was left unstandardized. The data were analyzed in two windows: days -10 to 0 , and days 0 to $+12$; these windows represent the conditioning period and the posttransplant/GVHD prophylaxis period, respectively. Hits were considered significant if they had a Benjamini-Hochberg-corrected P value less than 0.05 .

Some of the various computational tools that have been developed for determining microbial differential abundance between groups have been observed to produce different sets of results under certain circumstances. To validate the bivariate results of MaAsLin2, we also used Corncob (23), ANCOM2 (24), ANCOM-BC (25), and metagenomeSeq (Bioconductor package; refs. 26–28) to assess the compositional data from this study under various statistical approaches. Of these tools, only MaAsLin2 and ANCOM2 natively support the analysis of repeated measures from the same subjects.

Healthy volunteers

Healthy volunteers who provided stool samples provided written informed consent according to a biospecimen-collection protocol approved by the Memorial Sloan Kettering Cancer Center IRB.

Data availability statement

Data sharing requests can be e-mailed to the corresponding author.

Results

Patients

Of 1,903 allo-HCT recipients, 1,188 met the inclusion criteria for this observational cohort; key among these criteria were that the patient received one relatively common conditioning regimen and had donated an evaluable fecal specimen between days -20 and $+30$ relative to transplantation (**Fig. 1A**; **Table 1**). CD34-selected peripheral blood stem cells (PBSC) were the most common graft source [498 (41.9%)], followed by unmodified PBSC [428 (36.0%)]. Population and transplantation features differed between the

conditioning regimens (Supplementary Fig. S1; Supplementary Table S1). Most patients with lymphoma and chronic lymphocytic leukemia were conditioned with fludarabine, cyclophosphamide, and TBI 200 cGy (Flu/Cy/TBI200; ref. 29), the only NMA regimen. Busulfan, melphalan, and fludarabine (Bu4/Mel/Flu), a MAC regimen frequently used before infusion of CD34-selected grafts, was commonly administered in those with myeloid malignancies and myeloma. Recipients of CD34-selected transplants did not receive any additional agents for GVHD prophylaxis. Unmodified PBSC grafts and methotrexate-based GVHD prophylaxis were most common after fludarabine and busulfan at a MAC dose (Flu/Bu4), fludarabine and melphalan (Flu/Mel), and Flu/Cy/TBI200 conditioning. Exposure to nonprophylactic antibiotics (i.e., antibiotics initiated empirically, as for fever or a documented infection) between days 0 and 21 was highest with TBI1375.

Factors other than antibiotics contribute to changes in the microbiome of allo-HCT recipients

To characterize patterns of microbiota injury, we used 16S rRNA gene sequencing to analyze 7,682 stool samples (mean of 6.5 samples per patient) collected between days -20 and $+30$ (Supplementary Fig. S2). Alpha-diversity, a measure that reflects the number of unique bacteria present and their relative frequencies, was measured by the inverse Simpson index. Consistent with our previous studies (11), α -diversity markedly decreased during allo-HCT; in particular, the steepest decline was observed early in the treatment course, especially before day 0 (Fig. 1C, black curve; Supplementary Fig. S2). The reduction was only partially explained by exposure to nonprophylactic antibiotics and any antibacterial antibiotics (Fig. 1C, blue and purple curves). Among samples collected before graft infusion and not exposed to any type of bacterial antibiotics, diversity was lower in samples collected between days -5 and 0 (Fig. 1C, purple line, period t_2) than those collected between days -20 and -6 (Fig. 1C, purple line, period t_1 ; Fig. 1D; $P < 0.01$). Furthermore, in samples not exposed to antibacterial antibiotics, the similarity of the patient samples and healthy controls (measured as the Bray–Curtis distance to the average of the healthy-volunteer samples) still increased over time (Fig. 1E, $P < 0.001$). Therefore, additional factors besides antibiotics must be invoked to explain gut-microbiome disruption in allo-HCT recipients.

Conditioning regimen-associated microbiota injury

Stool consistency is a surrogate for microbiome composition (30). The fraction of stool samples with liquid consistency (Fig. 1F) correlated with conditioning intensity (MAC 72%, RIC 61%, NMA 27%, $P < 0.001$), suggesting microbiome injury is highest with MAC conditioning. Diversity declined over time in recipients of all three conditioning intensities (Fig. 2A). However, the decrease was greatest in recipients of MAC, followed by RIC and NMA (Supplementary Table S2). Notably, NMA was also associated with preservation of diversity compared with MAC in an external, previously reported (11) multicenter cohort ($n = 291$; Supplementary Table S3). Therefore, conditioning intensity partially explains the variance in diversity trajectories among allo-HCT recipients.

We hypothesized that microbiota injury varies not only by conditioning intensity category but also between specific conditioning regimens. Patient samples followed a similar overall pattern of diversity reduction regardless of intensity (Fig. 2B). However, diversity loss in samples collected shortly after graft infusion was

most pronounced with TBI 1,375 cGy, thiotepea, and cyclophosphamide (TBI1375/Tt/Cy) and fludarabine, cyclophosphamide, thiotepea, and TBI 400 cGy (Flu/Cy/Tt/TBI400) and lowest with Flu/Cy/TBI200 (Fig. 2C).

We next asked whether conditioning regimens induce specific taxonomic changes in the fecal microbiome. Microbial composition shifted over time with all regimens, as measured by Bray–Curtis distance: compared with preconditioning samples, recipients of TBI135/Tt/Cy and clofarabine, melphalan, and thiotepea (Clo/Tt/Mel) exhibited the greatest magnitude of change, and recipients of Flu/Cy/TBI200 the least (Fig. 2D). Nonetheless, the cumulative incidence of bacterial monodominance (defined as a relative abundance threshold of any single ASV of $\geq 30\%$; ref. 11) exceeded 80% by day $+30$, irrespective of the conditioning regimen (Supplementary Fig. S3). To explore the specific taxonomic changes that underpinned these shifts in global composition, we constructed a MaAsLin2 multivariable linear model (22) in which the outcome variables were the changes in genus relative abundances and the exposure variables were time relative to transplantation and exposure to each of the conditioning regimens (Fig. 2E). The red and blue circles in the figure indicate relative enrichment or depletion of taxa between days 0 and 12, respectively; the size and intensity of the circles convey the magnitude of change and statistical significance, respectively. For example, a relative decline in *Blautia* abundance was associated with all the regimens except Flu/Cy/TBI200. Furthermore, Flu/Cy/TBI200 had a pattern distinct from the others with relative preservation of members of the *Clostridia* class, including *Anaerostipes* and *Ruminococcus*; an inverse relationship was seen with many of the other regimens and *Clostridia*. To validate the bivariate results of MaAsLin2, we used additional modeling tools that rely on different statistical approaches (Supplementary Fig. S4; refs. 23–28). Overall, there was strong agreement between the methods.

In a similar MaAsLin2 model also taking into account antibiotic exposures, underlying disease, GVHD prophylaxis, and other clinical features, a preponderance of the significant associations between taxa and clinical variables were observed for the antibiotic exposures. Oral vancomycin and piperacillin–tazobactam were associated with extensive taxonomic shifts, including a reduction in many of the commensal bacteria (Supplementary Fig. S5A). Nevertheless, conditioning-associated microbiome changes were still evident in the multivariable models, despite a correlation between conditioning regimens and antibiotic exposures (Supplementary Fig. S6). For instance, many taxonomic shifts were observed in the Flu/Cy/Tt/TBI400 regimen including the loss of *Blautia*, *Ruminococcus*, and other *Ruminococcaeae*. GVHD prophylaxis was associated with a significant reduction of *Clostridia*, including *Blautia*, *Anaerostipes*, *Dorea*, and *Drancourtella*. We therefore constructed an additional multivariable model separating GVHD prophylaxis to individual elements (Supplementary Fig. S5B). Methotrexate and cyclophosphamide, which were used in mutually exclusive fashion for GVHD prophylaxis (Supplementary Fig. S6), were both associated with *Clostridia* loss, although numerically a greater change was observed with methotrexate. In contrast, cyclosporine and mycophenolate mofetil were associated with minimal microbial changes. Notably, in analysis of individual antibiotic or conditioning drug exposure in the preinfusion time window, antibiotic exposures had many strong associations (Supplementary Fig. S5C). Oral vancomycin was associated with the greatest shifts. Collectively, our findings support a differential association between conditioning regimens and microbiome composition.

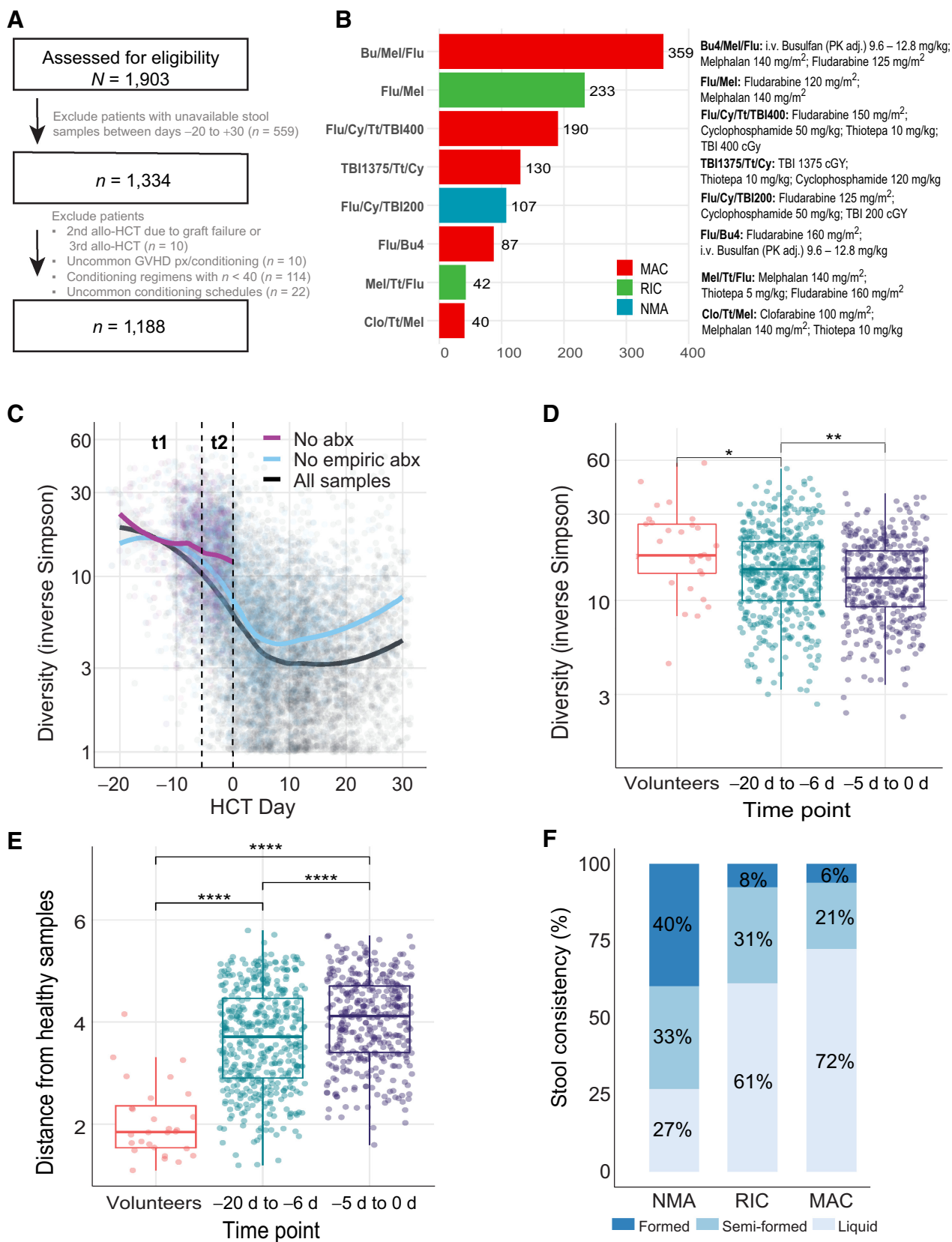


Table 1. Population characteristics.

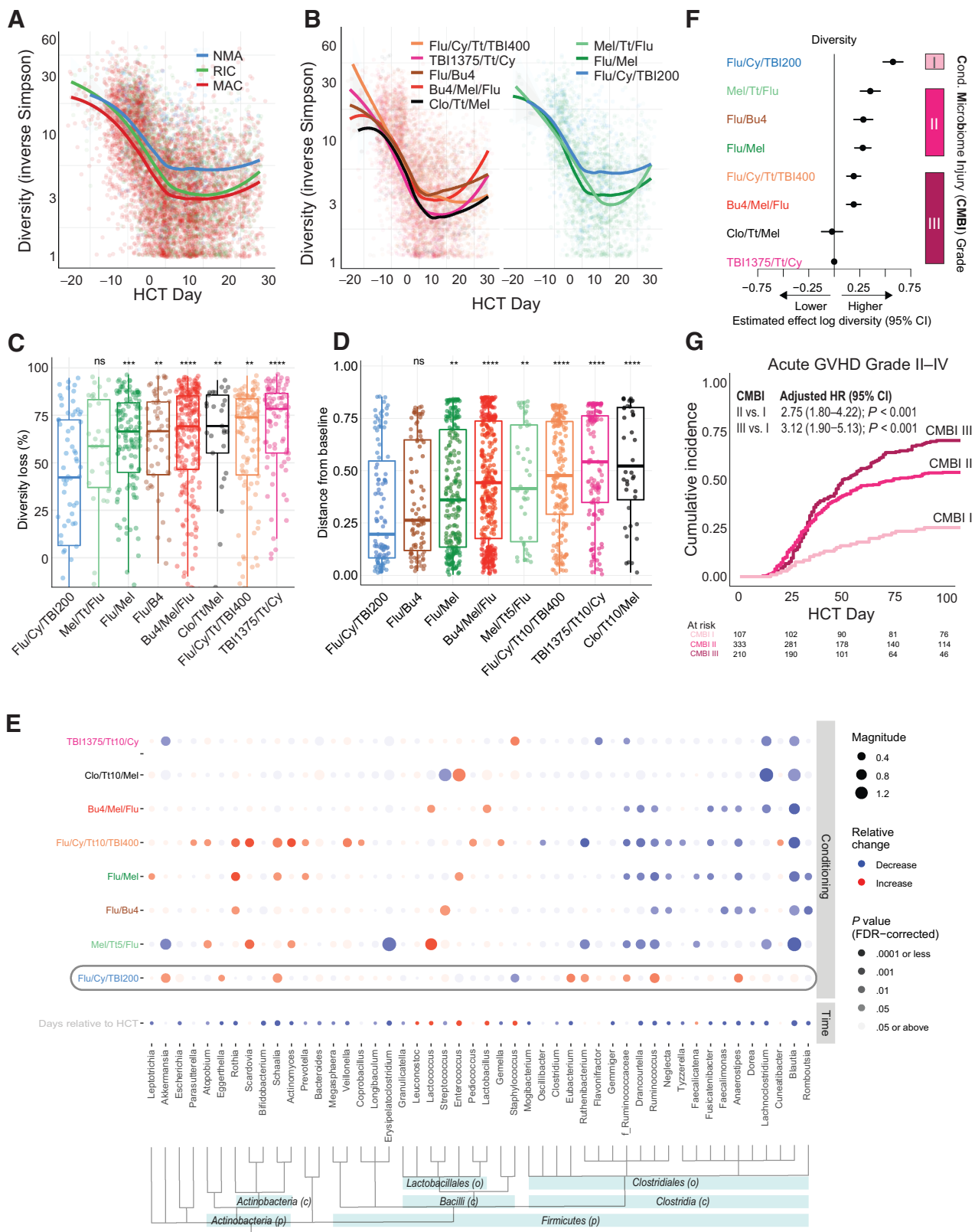
		Overall	Missing [n (%)]
<i>n</i>		1,188	
Sex (%)	Male	715 (60.2)	0
	Female	473 (39.8)	
Age [median (IQR)]		57 (47–64)	0
Disease (%)	Acute myeloid leukemia	418 (35.2)	0
	MDS or MPN	286 (24.1)	
	Lymphoma/CLL	231 (19.4)	
	ALL	110 (9.3)	
	Plasma cell neoplasm	111 (9.3)	
	Other leukemias	22 (1.9)	
	Nonmalignant disorders	10 (0.8)	
Previous allo-HCT (%)	First allo-HCT	1,141 (96.0)	0
	Second allo-HCT	47 (4.0)	
HCT-CI [median (IQR)]		2 (1–4)	25 (2.1%)
HCT year [median (IQR)]		2,015 (2,013–2,017)	0
Donor type (%)	Matched unrelated	534 (44.9)	0
	Matched related	307 (25.8)	
	Cord blood	174 (14.6)	
	Mismatched non-haploidentical	128 (10.8)	
	Haploidentical	45 (3.8)	
Graft source (%)	PBSC T-cell depleted	498 (41.9)	0
	PBSC unmodified	428 (36.0)	
	Cord blood	174 (14.6)	
	BM unmodified	88 (7.4)	
Conditioning intensity (%)	Ablative	806 (67.8)	0
	Reduced Intensity	275 (23.1)	
	Nonmyeloablative	107 (9.0)	
Conditioning regimens (%)	Bu4/Mel/Flu	359 (30.2)	0
	Flu/Mel	233 (19.6)	
	Flu/Cy/Tt/TBI400	190 (16.0)	
	TBI1375/Tt/Cy	130 (10.9)	
	Flu/Bu4	87 (7.3)	
	Flu/Cy/TBI200	107 (9.0)	
	Mel/Tt/Flu	42 (3.5)	
	Clo/Tt/Mel	40 (3.4)	
GVHD prophylaxis (%)	None	498 (41.9)	0
	MTX-based	426 (35.9)	
	MMF-based	186 (15.7)	
	PTCy-based	78 (6.6)	
Exposure to antibiotics ^a up to day 21 (%)	Exposed	804 (67.7)	0
	Not exposed	384 (32.3)	

Abbreviations: ALL, acute lymphoblastic leukemia; allo-HCT, allogeneic hematopoietic cell transplantation; AML, acute myeloid leukemia; CLL, chronic lymphocytic leukemia; HCT-CI, hematopoietic cell transplantation-specific comorbidity index; IQR, interquartile range; MDS, myelodysplastic syndrome; MMF, mycophenolate mofetil; MPN, myeloproliferative neoplasm; MTX, methotrexate; PBSC, peripheral blood stem cells; PTCy, post-transplantation cyclophosphamide.

^aThe frequency of antibiotic exposure considered any exposure to drugs commonly used in this cohort for treatment of neutropenic fever or for *C. difficile* diarrhea (oral vancomycin, imipenem-cilastatin, meropenem, piperacillin-tazobactam, clindamycin, and metronidazole) between the first day of conditioning and day 21 post-allo-HCT.

Figure 1.

Fecal samples from 1,188 recipients of allo-HCT were analyzed. **A**, CONSORT diagram for patient inclusion. The patients were conditioned with one of the eight regimens received by >40 patients with at least one stool sample collected between days –20 and +30. **B**, Flu/Cy/TBI200 was the only NMA regimen. Flu/Mel and Mel/Tt/Flu were the RIC conditioning regimens; the rest of the regimens were MAC. **C**, Not all diversity loss can be attributed to antibiotic exposures. Each point is a stool sample whose α -diversity (as measured by 16S amplicon sequencing and the inverse Simpson index) is plotted over time relative to transplantation. Lines are smoothed average by the LOESS (locally estimated scatterplot smoothing) method, in which each data point (α -diversity measurement) is considered as an independent event. The black curve averages all samples, including those collected before or after any antibiotic exposure. The purple curve ignores any sample collected after exposure to any antibacterial antibiotic; as virtually all patients commenced prophylactic antibiotics by day –2 or with the onset of neutropenia, the purple curve does not extend beyond day 0. The blue curve ignores any samples collected after exposure to nonprophylactic antibiotics, that is, samples exposed to the prophylactic antibiotics ciprofloxacin and intravenous vancomycin are included in the blue smoothed average. Time bin *t*1 (days –20 to –6) and *t*2 (days –5 to 0) are indicated with dashed vertical lines. **D**, Among samples from *n* = 507 patients whose samples were not exposed to any antibacterial antibiotics, α -diversity declines significantly between *t*1 and *t*2, indicating that not all diversity loss can be attributed to antibiotic exposures. *t*1 α -diversity is also significantly lower than those of healthy volunteers (*n* = 30). Only one stool sample per patient was considered in each time window. **E**, Among samples from *n* = 507 patients whose samples were not exposed to any antibacterial antibiotics, microbiota composition is significantly different between *t*1 and *t*2, as measured by Bray–Curtis distance to a centroid of healthy volunteer (*n* = 28) compositions, indicating that not all composition changes can be attributed to antibiotic exposures. **F**, Liquid stool samples (*n* = 957), collected between days 0 and +10, were more commonly collected from recipients of more intense conditioning regimens, as assessed by laboratory technicians at the time of sample aliquoting. X-squared for all samples = 118.44, df = 4, $P < 2.2 \times 10^{-16}$; *, $P \leq 0.05$; **, $P \leq 0.01$; ***, $P \leq 0.001$; ****, $P \leq 0.0001$. Abx, antibiotics; allo-HCT, allogeneic hematopoietic cell transplantation; i.v., intravenous; MAC, myeloablative conditioning; NMA, nonmyeloablative conditioning; RIC, reduced-intensity conditioning.



Conditioning-associated microbiota injury grading

To provide a scale quantifying the degree of microbiota injury by regimen, we constructed a generalized estimating equation multivariable model for α -diversity and then manually grouped the resulting diversity-reduction coefficients into three levels of Conditioning-associated Microbiome Injury (CMBI; Fig. 2F; Supplementary Table S4): low (CMBI I: Flu/Cy/TBI200), intermediate [CMBI II: Flu/Mel, Flu/Bu4, Mel/Tt/Flu (melphalan, thiotepe, and fludarabine)], and high (CMBI III: Flu/Cy/TBI400, Bu4/Mel/Flu, Clo/Tt/Mel, TBI1375/Tt/Cy). Notably, the MAC regimen Flu/Bu4 (1, 2) was associated with intermediate microbiota toxicity. Among recipients of unmanipulated grafts (i.e., excluding CD34-selected), regimens assigned to intermediate- and high-grade CMBI had a higher risk of grade II–IV acute GVHD compared with low-grade MBI (Fig. 2G). Altogether, our findings indicate that microbiota injury patterns are regimen specific and can serve as a clinical biomarker.

Discussion

This observational study is the first to show regimen-specific relationships between individual conditioning regimens and fecal microbiome composition. Conditioning-associated changes in the microbiome were observed with global metrics of microbiome composition, such as α - and β -diversity, as well as at the taxonomic level. Multivariable modeling illustrated that antibiotic exposures and GVHD prophylaxis also contribute to microbiome perturbations after HCT. Our results in patients with cancer extend prior reports of interactions between nonantibiotic drugs and gut microbiome in healthy volunteers, inflammatory bowel disease, and *in vitro* analyses (17, 31, 32).

Conditioning regimens are the backbone of allogeneic HCT. Conditioning intensity categories (1, 2) reflect the tradeoff between the degree of tumor eradication and toxicity, both being greatest with MAC regimens and reduced with NMA regimens. We propose microbiota injury as a new axis for evaluating conditioning regimens and provide a classification of commonly used regimens that summarizes their degree of microbiota toxicity, which may be relevant to clinical outcomes such as GVHD and immune reconstitution (9, 11, 33–35). In contrast to intensity classification, which was based on expert opinion (1, 2, 36), we used a data-driven approach to quantitate microbiota changes and define injury levels. We defined injury based on α -diversity because it is a measure associated with survival after allo-HCT (11, 37). Given this association, there

are ongoing efforts to increase diversity in allo-HCT recipients (38, 39). Our results suggest that conditioning regimens categorized with high-grade CMBI may serve as the optimal candidates for such interventions.

Nonantibiotic drugs can influence growth of bacterial population (17, 31, 32). Our observation that changes in microbial patterns are conditioning-regimen dependent could be explained by several potential mechanisms. First, there could be direct effects of chemotherapy or radiation on bacteria. Indeed, radiation and cyclophosphamide, which are both commonly used as part of the conditioning backbone, have been shown to induce microbial changes in mice (18, 19, 40). Conditioning may also lead to bacterial changes indirectly. Chemotherapy and radiation damage the mucosal membrane and elements of the intestinal epithelium, such as Paneth cells that secrete antibacterial peptides (41–44).

Isolating the impact of conditioning regimens on the microbiota is challenging because confounders such as antibiotics, GVHD prophylaxis, and nutrition also affect bacterial populations. Nonetheless, we observed associations between conditioning regimens and microbiome perturbation, even after accounting for antibiotic and GVHD prophylaxis drug exposures. There are additional observations from the literature supporting conditioning as a driver of microbiome shifts. Maier and colleagues demonstrated that nonantibiotic drugs, including chemotherapeutic drugs, have an inhibiting effect on bacterial growth *in vitro* (17). Others have shown that radiation and cyclophosphamide (both commonly used for conditioning) have an impact on the gut microbiome (6, 18, 19, 40). The multivariable models illustrate the profound impact of antibiotics on the microbiome. Specifically, oral vancomycin and intravenous piperacillin–tazobactam were associated with dramatic changes in the microbiome, including butyrate-producing members of the Clostridiaceae family. Our findings raise the possibility that antibiotic stewardship and rational selection of antibiotics could substantially mitigate microbiota injury in allo-HCT recipients, a hypothesis we are testing in a randomized trial (NCT03078010).

Our findings indicate that conditioning regimens contribute to microbiota injury in allo-HCT recipients. Notably, the degree of injury is not directly correlated to classical conditioning intensity levels. For instance, conditioning with fludarabine and busulfan at a MAC dose (Flu/Bu4) was associated with intermediate microbiota injury (CMBI II). Because conditioning regimens are modifiable, there may be a rationale to select less-microbiome-disruptive regimens in certain patients, perhaps those at higher risk of GVHD or those with already

Figure 2.

Microbiota injury patterns in allo-HCT are conditioning regimen specific. **A**, Fecal microbiota α -diversity declines most in recipients of MAC and the least in recipients of NMA conditioning (Supplementary Table S2). Each point is a stool sample whose α -diversity is plotted over time relative to transplantation. The smoothed curves plots are moving averages of recipients of different regimens. **B**, Distinct patterns of α -diversity dynamics can be observed among recipients of specific conditioning regimens. Five MAC regimens are plotted in the left; two RIC and one NMA regimens are plotted in the right. **C**, In a patient-level pair-matched analysis of the earliest sample collected before (days –20 and –1) and earliest sample after conditioning (days 0 to +10), the NMA regimen Flu/Cy/TBI200 was associated with the least reduction in diversity (42%). In contrast, the MAC regimens were associated with the greatest reduction (69%–78%). Regimens were compared with Flu/Cy/TBI200 with the Wilcoxon signed-rank test. **D**, Microbiome dissimilarity, as determined by applying multidimensional scaling to the Bray–Curtis distance matrix, between samples collected at days 0 and +12 was significantly increased in all conditioning regimens except Flu/Cy/TBI200. The greatest distance was in the MAC regimen Clo/Tt/Mel. **E**, A multivariable MaAsLin2 model (22) adjusting for time of sampling and conditioning regimens reveals a differential association between microbial taxa and conditioning regimens. Bacteria are ordered by taxonomical ranking. ASVs that could not be classified to the genus level were analyzed at the family level, as indicated by *f_Ruminococcaceae*. Flu/Cy/TBI200, highlighted with a gray rounded rectangle, was associated with the preservation of members of the Clostridia class. **F**, A grading scheme classifying regimens into three categories of microbiota injury [low (I), intermediate (II), and high (III)] based diversity reduction between days –20 and +30 was introduced. The estimated effect on diversity of each regimen was derived from a generalized estimating equation (Supplementary Table S4) regression model, adjusting for time, age, sex, exposure to GVHD prophylaxis, antibiotics, and conditioning. **G**, Patients who received unmodified grafts had a higher incidence of acute GVHD grade II–IV with regimens categorized as having greater microbiota toxicity. CMBI HRs are derived from a multivariable Cox-regression adjusting for age, sex, comorbidity burden, donor and HLA matching, and GVHD prophylaxis. *, $P \leq 0.05$; **, $P \leq 0.01$; ***, $P \leq 0.001$; ****, $P \leq 0.0001$. CMBI, conditioning-associated microbiota injury; HCT, allogeneic hematopoietic cell transplantation; MAC, myeloablative conditioning; NMA, nonmyeloablative conditioning; RIC, reduced-intensity conditioning.

injured microbiome composition at baseline. Finally, the microbiome injury index could potentially be used for risk stratification in clinical trials studying microbiome-based interventions in HCT.

Authors' Disclosures

R. Shouval reports personal fees from Medexus and MyBiotics outside the submitted work. N.R. Waters reports Memorial Sloan Kettering Cancer Center has financial interests relative to Seres Therapeutics. A.L.C. Gomes reports other support from Xbiome Corp. outside the submitted work. K.A. Markey reports other support from Postbiotics Plus and Incyte outside the submitted work. A.G. Clurman reports grants from NIH/NCI during the conduct of the study. T.M. Hohl reports grants from NIH during the conduct of the study, as well as other support from Boehringer Ingelheim outside the submitted work. A.D. Sung reports grants from American Society of Hematology and Damon Runyon Cancer Research Foundation during the conduct of the study; A.D. Sung also reports non-financial support from DSM/iHealth and Clasado, as well as grants from Merck and Novartis outside the submitted work. D. Hashimoto reports personal fees from Novartis Pharma KK, Janssen Pharmaceutical KK, Eisai Co., Ltd., Chugai Pharmaceutical Co., Ltd., Ono Pharmaceutical Co., Ltd., Sumitomo Pharma Co., Ltd., and Merck & Co., Inc. outside the submitted work. T. Teshima reports grants and personal fees from Chugai, Kyowa Kirin, and Nippon Shinyaku; grants from Fuji Pharma, Eisai, Ono, Shionogi, Priothera SAS, LUCA Science, and Otsuka; grants and non-financial support from Asahi Kasei Pharma; grants, personal fees, and non-financial support from Sumitomo Pharma and Astellas; personal fees from AbbVie, Bristol Myers Squibb, and Merck Sharp & Dohme; personal fees and non-financial support from Novartis, Celgene, and Janssen; and non-financial support from Meiji Seika Pharma, Daiichi Sankyo, AstraZeneca, Takeda, Roche Diagnostics, and Sanofi outside the submitted work. E. Holler reports personal fees from PharmaBiome (Switzerland), MaaT Pharma (France), Novartis, and Neovii, as well as grants from Medac (Germany) outside the submitted work. M. Scordo reports personal fees and other support from Angiocrine Bioscience, Inc. and Omeros Corporation; other support from Amgen, Inc.; and personal fees from Kite—A Gilead Company, i3Health, Medscape, LLC, and McKinsey & Company outside the submitted work. S.A. Giralat reports other support from Amgen, BMS, Sanofi, Pfizer, Janssen, GSK, Jazz, Celgene, and Actinnum outside the submitted work. M.-A. Perales reports personal fees from Adicet, Allovir, Caribou Biosciences, Syncopation, Bristol Myers Squibb, VectivBio AG, Equilibrium, Exevir, Karyopharm, Miltenyi Biotec, MorphoSys, Merck, Omeros, OrcaBio, and NexImmune, as well as personal fees and other support from Vor Biopharma, Novartis, Incyte, Kite/Gilead, and Nektar Therapeutics outside the submitted work. J.U. Peled reports grants from NHLBI NIH Award K08HL143189, P30 CA008748, Parker Institute for Cancer Immunotherapy at Memorial Sloan Kettering, and American Society for Transplantation and Cellular Therapy during the conduct of the study. J.U. Peled also reports personal fees, non-financial support, and other support from Seres Therapeutics; personal fees from DaVolterra, CSL Behring, and MaaT Pharma; and personal fees and other support from Postbiotics Research outside the submitted work. In addition, J.U. Peled has a patent for methods and compositions for detecting risk of cancer relapse pending and licensed to Seres Therapeutics, a patent for bacterial compositions and methods for cancer survival pending and licensed, and a patent for beta-lactamase compositions for treatment of graft versus host disease pending. Memorial Sloan Kettering Cancer Center has financial interests relative to Seres Therapeutics. M.R.M. van den Brink reports other support from Seres, Vor Biopharma, Rheos, Frazier Healthcare Partners, Nektar Therapeutics, Notch Therapeutics, Ceramedix, Lygenesis, Pluto Therapeutics, GSK, Da Volterra, Thymofox, Garuda, Wolters Kluwer, Juno, DKMS, Smart Immune, and Thymofox outside the submitted work. No disclosures were reported by the other authors.

References

1. Bacigalupo A, Ballen K, Rizzo D, Giralat S, Lazarus H, Ho V, et al. Defining the intensity of conditioning regimens: working definitions. *Biol Blood Marrow Transplant* 2009;15:1628–33.
2. Giralat S, Ballen K, Rizzo D, Bacigalupo A, Horowitz M, Pasquini M, et al. Reduced-intensity conditioning regimen workshop: defining the dose spectrum. Report of a workshop convened by the center for international blood and marrow transplant research. *Biol Blood Marrow Transplant* 2009;15:367–9.

Authors' Contributions

R. Shouval: Conceptualization, data curation, software, formal analysis, funding acquisition, investigation, visualization, methodology, writing—original draft, project administration, writing—review and editing. **N.R. Waters:** Formal analysis, visualization, methodology, writing—review and editing. **A.L.C. Gomes:** Formal analysis, methodology, writing—review and editing. **C. Zuanelli Brambilla:** Software, formal analysis, writing—review and editing. **T. Fei:** Software, methodology, writing—review and editing. **S.M. Devlin:** Software, formal analysis, methodology, writing—review and editing. **C.L. Nguyen:** Validation, writing—review and editing. **K.A. Markey:** Validation, writing—review and editing. **A. Dai:** Software, writing—review and editing. **J.B. Slingerland:** Data curation. **A.G. Clurman:** Data curation. **E. Fontana:** Data curation. **L.A. Amoretti:** Data curation. **R.J. Wright:** Data curation. **T.M. Hohl:** Resources, writing—review and editing. **Y. Taur:** Resources, software. **A.D. Sung:** Resources, data curation, writing—review and editing. **D. Weber:** Resources, data curation, writing—review and editing. **D. Hashimoto:** Resources, data curation, writing—review and editing. **T. Teshima:** Resources, data curation, writing—review and editing. **N.J. Chao:** Resources, data curation, writing—review and editing. **E. Holler:** Resources, data curation, writing—review and editing. **M. Scordo:** Resources, writing—review and editing. **S.A. Giralat:** Resources, writing—review and editing. **M.-A. Perales:** Resources, writing—review and editing. **J.U. Peled:** Conceptualization, resources, software, formal analysis, supervision, funding acquisition, validation, investigation, methodology, writing—original draft, writing—review and editing. **M.R.M. van den Brink:** Conceptualization, resources, supervision, funding acquisition, project administration, writing—review and editing.

Acknowledgments

This research was supported by NCI award numbers, R01-CA228358, R01-CA228308, P30 CA008748 MSK Cancer Center Support Grant/Core Grant and P01-CA023766; National Heart, Lung, and Blood Institute (NHLBI) award number R01-HL123340 and R01-HL147584; National Institute of Aging award number P01-AG052359; and Tri Institutional Stem Cell Initiative. Additional funding was received from The Lymphoma Foundation, The Susan and Peter Solomon Divisional Genomics Program, Cycle for Survival, and the Parker Institute for Cancer Immunotherapy.

T.M. Hohl reports support from NIAID NIH Award R21AI156157.

J.U. Peled reports funding from NHLBI NIH Award K08HL143189.

R. Shouval was supported by the American Society of Transplantation and Cellular Therapy New Investigator Award, the American Society of Hematology Fellow Scholar Award, a grant from the Long Island Sound Chapter, Swim Across America, the Robert Hirschhorn Award, and the Memorial Sloan Kettering Steven Greenberg Lymphoma Research Award.

K.A. Markey would like to acknowledge funding from the DKMS and American Society for Hematology, the Royal Australasian College of Physicians, The Haematology Society of Australia and New Zealand, and the American Australian Association.

The publication costs of this article were defrayed in part by the payment of publication fees. Therefore, and solely to indicate this fact, this article is hereby marked “advertisement” in accordance with 18 USC section 1734.

Note

Supplementary data for this article are available at Clinical Cancer Research Online (<http://clincancerres.aacrjournals.org/>).

Received April 19, 2022; revised September 13, 2022; accepted October 31, 2022; published first November 2, 2022.

3. Cooke KR, Luznik L, Sarantopoulos S, Hakim FT, Jagasia M, Fowler DH, et al. The biology of chronic graft-versus-host disease: a task force report from the National Institutes of Health consensus development project on criteria for clinical trials in chronic graft-versus-host disease. *Biol Blood Marrow Transplant* 2017;23:211–34.
4. Shouval R, Fein JA, Shouval A, Danylesko I, Shem-Tov N, Zlotnik M, et al. External validation and comparison of multiple prognostic scores in allogeneic hematopoietic stem cell transplantation. *Blood Advances* 2019;3:1881–90.

5. Litvak Y, Byndloss MX, Baumler AJ. Colonocyte metabolism shapes the gut microbiota. *Science* 2018;362:eaat9076.
6. Jenq RR, Ubeda C, Taur Y, Menezes CC, Khanin R, Dudakov JA, et al. Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J Exp Med* 2012;209:903–11.
7. Hansson GC, Johansson ME. The inner of the two Muc2 mucin-dependent mucus layers in colon is devoid of bacteria. *Gut Microbes* 2010;1:51–4.
8. Levy M, Thaiss CA, Zeevi D, Dohnalová L, Zilberman-Schapira G, Mahdi JA, et al. Microbiota-modulated metabolites shape the intestinal microenvironment by regulating NLRP6 inflammasome signaling. *Cell* 2015;163:1428–43.
9. Stein-Thoeringer CK, Nichols KB, Lazrak A, Docampo MD, Slingerland AE, Slingerland JB, et al. Lactose drives *Enterococcus* expansion to promote graft-versus-host disease. *Science* 2019;366:1143–9.
10. Shono Y, van den Brink MR. Gut microbiota injury in allogeneic haematopoietic stem cell transplantation. *Nat Rev Cancer* 2018;18:283–95.
11. Peled JU, Gomes AL, Devlin SM, Littmann ER, Taur Y, Sung AD, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med* 2020;382:822–34.
12. Stoma I, Littmann ER, Peled JU, Giralto S, van den Brink MR, Pamer EG, et al. Compositional flux within the intestinal microbiota and risk for bloodstream infection with gram-negative bacteria. *Clin Infect Dis* 2021; 73:e4627–e35.
13. Morjaria S, Schluter J, Taylor BP, Littmann ER, Carter RA, Fontana E, et al. Antibiotic-induced shifts in fecal microbiota density and composition during hematopoietic stem cell transplantation. *Infect Immun* 2019;87:e00206–19.
14. Rashidi A, Kaiser T, Holtan SG, Rehman TU, Weisdorf DJ, Khoruts A, et al. Levaquin gets a pass. *Biol Blood Marrow Transplant* 2020;26:778–81.
15. Shono Y, Docampo MD, Peled JU, Perobelli SM, Velardi E, Tsai JJ, et al. Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice. *Sci Transl Med* 2016;8:339ra71.
16. Holler E, Butzhammer P, Schmid K, Hundsruker C, Koestler J, Peter K, et al. Metagenomic analysis of the stool microbiome in patients receiving allogeneic stem cell transplantation: loss of diversity is associated with use of systemic antibiotics and more pronounced in gastrointestinal graft-versus-host disease. *Biol Blood Marrow Transplant* 2014;20:640–5.
17. Maier L, Pruteanu M, Kuhn M, Zeller G, Telzerow A, Anderson EE, et al. Extensive impact of non-antibiotic drugs on human gut bacteria. *Nature* 2018; 555:623–8.
18. Guo H, Chou WC, Lai Y, Liang K, Tam JW, Brickley WJ, et al. Multi-omics analyses of radiation survivors identify radioprotective microbes and metabolites. *Science* 2020;370:eaay9097.
19. Crawford PA, Gordon JL. Microbial regulation of intestinal radiosensitivity. *Proc Natl Acad Sci U S A* 2005;102:13254–9.
20. Javdan B, Lopez JG, Chankhamjon P, Lee YJ, Hull R, Wu Q, et al. Personalized mapping of drug metabolism by the human gut microbiome. *Cell* 2020;181: 1661–79.
21. Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJ, Holmes SP. DADA2: high-resolution sample inference from Illumina amplicon data. *Nat Methods* 2016;13:581–3.
22. Mallick H, Rahnavard A, McIver LJ, Ma S, Zhang Y, Nguyen LH, et al. Multivariable association discovery in population-scale meta-omics studies. *PLoS Comput Biol* 2021;17:e1009442.
23. Martin BD, Witten D, Willis AD. Modeling microbial abundances and dysbiosis with beta-binomial regression. *Ann Appl Stat* 2020;14:94–115.
24. Kaul A, Mandal S, Davidov O, Peddada SD. Analysis of microbiome data in the presence of excess zeros. *Front Microbiol* 2017;8:2114.
25. Lin H, Peddada SD. Analysis of compositions of microbiomes with bias correction. *Nat Commun* 2020;11:3514.
26. Paulson JN, Stine OC, Bravo HC, Pop M. Differential abundance analysis for microbial marker-gene surveys. *Nat Methods* 2013;10:1200–2.
27. Paulson JN, Talukder H, Bravo HC. Longitudinal differential abundance analysis of microbial marker-gene surveys using smoothing splines. *bioRxiv*; 2017.
28. Paulson JN, Pop M, Bravo HC. metagenomeSeq: Statistical analysis for sparse high-throughput sequencing. *Bioconductor package*; 2013:191.
29. Sauter CS, Barker JN, Lechner L, Zheng J, Devlin SM, Papadopoulos EB, et al. A phase II study of a nonmyeloablative allogeneic stem cell transplant with peritransplant rituximab in patients with B cell lymphoid malignancies: favorably durable event-free survival in chemosensitive patients. *Biol Blood Marrow Transplant* 2014;20:354–60.
30. Vandeputte D, Falony G, Vieira-Silva S, Tito RY, Joossens M, Raes J, et al. Stool consistency is strongly associated with gut microbiota richness and composition, enterotypes and bacterial growth rates. *Gut* 2016;65:57–62.
31. Vich Vila A, Collij V, Sanna S, Sinha T, Imhann F, Bourgonje AR, et al. Impact of commonly used drugs on the composition and metabolic function of the gut microbiota. *Nat Commun* 2020;11:362.
32. Jackson MA, Verdi S, Maxan M-E, Shin CM, Zierer J, Bowyer RCE, et al. Gut microbiota associations with common diseases and prescription medications in a population-based cohort. *Nat Commun* 2018;9:2655.
33. Miltiadous O, Waters NR, Andriova H, Dai A, Nguyen CL, Burgos da Silva M, et al. Early intestinal microbial features are associated with CD4 T cell recovery after allogeneic hematopoietic transplant. *Blood* 2022;139:2758–69.
34. Markey KA, Schluter J, Gomes ALC, Littmann ER, Pickard AJ, Taylor BP, et al. The microbe-derived short-chain fatty acids butyrate and propionate are associated with protection from chronic GVHD. *Blood* 2020;136:130–6.
35. Schluter J, Peled JU, Taylor BP, Markey KA, Smith M, Taur Y, et al. The gut microbiota is associated with immune cell dynamics in humans. *Nature* 2020; 588:303–7.
36. Spyridonidis A, Labopin M, Savani BN, Niittyvuopio R, Blaise D, Craddock C, et al. Redefining and measuring transplant conditioning intensity in current era: a study in acute myeloid leukemia patients. *Bone Marrow Transplant* 2020;55: 1114–25.
37. Taur Y, Jenq RR, Perales MA, Littmann ER, Morjaria S, Ling L, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood* 2014;124:1174–82.
38. Van Lier YF, Van den Brink MR, Hazenberg MD, Markey KA. The post-hematopoietic cell transplantation microbiome: relationships with transplant outcome and potential therapeutic targets. *Haematologica* 2021;106:2042–53.
39. DeFilipp Z, Peled JU, Li S, Mahabamunuge J, Dagher Z, Slingerland AE, et al. Third-party fecal microbiota transplantation following allo-HCT reconstitutes microbiome diversity. *Blood Adv* 2018;2:745–53.
40. Viaud S, Saccheri F, Mignot G, Yamazaki T, Daillère R, Hannani D, et al. The intestinal microbiota modulates the anticancer immune effects of cyclophosphamide. *Science* 2013;342:971–6.
41. Salzman NH, Hung K, Haribhai D, Chu H, Karlsson-Sjöberg J, Amir E, et al. Enteric defensins are essential regulators of intestinal microbial ecology. *Nat Immunol* 2010;11:76–83.
42. Eriguchi Y, Takashima S, Oka H, Shimoji S, Nakamura K, Uryu H, et al. Graft-versus-host disease disrupts intestinal microbial ecology by inhibiting Paneth cell production of α -defensins. *Blood* 2012;120:223–31.
43. Weber D, Frauenschläger K, Ghimire S, Peter K, Panzer I, Hiergeist A, et al. The association between acute graft-versus-host disease and antimicrobial peptide expression in the gastrointestinal tract after allogeneic stem cell transplantation. *PLoS One* 2017;12:e0185265.
44. Levine JE, Huber E, Hammer ST, Harris AC, Greenson JK, Braun TM, et al. Low Paneth cell numbers at onset of gastrointestinal graft-versus-host disease identify patients at high risk for nonrelapse mortality. *Blood* 2013;122:1505–9.